

DANDI:001326: Brain-wide mouse electrophysiology during aversive stimuli and ketamine

Scope of this export (computed by this export): the units, electrodes and intervals tables of every NWB file were read remotely from the DANDI Archive; voltage, LFP, video and other time series were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in `dandi_export/` (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from `sub-0_ses-20230901_ecephys.nwb`.

Field	Value (DANDI metadata)
Name	Brain-wide mouse electrophysiology during aversive stimuli and ketamine
Identifier / version	DANDI:001326 / 0.250528.1957
Citation	Brain-wide mouse electrophysiology during aversive stimuli and ketamine (2025). (Version 0.250528.1957) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.001326/0.250528.1957
DOI	10.48324/dandi.001326/0.250528.1957
Contributors (roles)	National Institutes of Health (NIH) (Funder); Richman, Ethan (Author, ContactPerson); Kauvar, Isaac (Author)
License	CC-BY-4.0
Keywords	Neuropixels, Mouse, Electrophysiology, Ketamine, Timescale
Related resources	Analysis code (IsSupplementTo)
Files / size / subjects	13 files, 2.61 GB, 10 subjects
Species	Mus musculus - House mouse
Approach	electrophysiological approach
Measurement technique	spike sorting technique; surgical technique
Variables measured	ElectrodeGroup, Units
Date published	2025-05-28T19:57:47.043354+00:00

Description (DANDI metadata, verbatim)

Brain-wide electrophysiology associated with Kauvar*, Richman*, Liu* et al. Science (2025). <https://doi.org/10.1126/science.adt3971>

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	Mouse extracellular ephys data for Kauvar*, Richman*, Liu* et al. Science (2025)
experiment_description	Neuropixels 1.0 recording with eyepuff assay and ketamine administration.
institution	Stanford University
lab	Deisseroth
experimenter	Ethan Richman, Chelsea Li
keywords	neuropixels; extracellular electrophysiology; affective behavior; emotion-like state; ketamine; eyepuff; air puff
related_publications	Kauvar*, Richman*, Liu*, et al., Conserved brain-wide emergence of emotional response from sensation in humans and mice. Science (2025) [DOI TBD]
subject	subject_id: 0; species: Mus musculus; strain: C57Bl/6J; genotype: wt; sex: F; age: P8W/P12W; description: mouse
devices	Neuropixels_Probe_0: Neuropixels 1.0 probe; Neuropixels_Probe_1: Neuropixels 1.0 probe; Neuropixels_Probe_2: Neuropixels 1.0 probe; Neuropixels_Probe_3: Neuropixels 1.0 probe

Field	Value (NWB file)
electrode_groups	ElectrodeGroup0: description = 'Probe 0 electrode group. xyz is in AIBSATlas coordinate space'; ElectrodeGroup1: description = 'Probe 1 electrode group. xyz is in AIBSATlas coordinate space'; ElectrodeGroup2: description = 'Probe 2 electrode group. xyz is in AIBSATlas coordinate space'; ElectrodeGroup3: description = 'Probe 3 electrode group. xyz is in AIBSATlas coordinate space'

2. Contents of the files

Quantity	Value	Source
NWB files read / subjects	13 / 10	NWB file
Species	Mus musculus (13)	NWB file
Sex (files)	F (13)	NWB file
Genotype (files)	wt (13)	NWB file
Strain (files)	C57Bl/6J (13)	NWB file
Age (files)	P8W/P12W (13)	NWB file
Lab / institution	Deisseroth (13) / Stanford University (13)	NWB file
Units (total)	9,554	NWB file
Units per file	median 851 (range 154–1020)	NWB file
Spikes (sum of spike_times_index lengths)	324,856,273	NWB file
How region was assigned to units	units.brain_region	Computed by this export
Distinct region labels	150	NWB file
units.quality values	good: 9,554	NWB file

The meaning of each units column, including any quality label, is quoted from the NWB file in section 7.

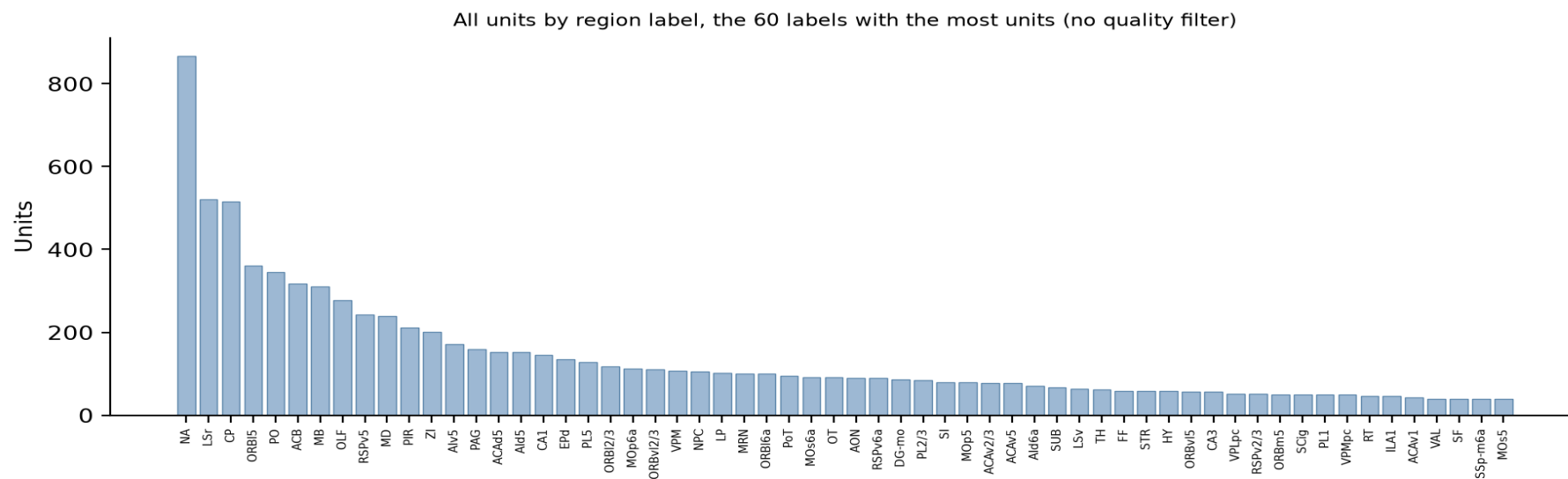
3. NWB files

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-0_ses-20230901_ecephys	0	F	P8W/P12 W	0.24	942	1506	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(952)
sub-1_ses-20230218_ecephys	1	F	P8W/P12 W	0.14	583	1530	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(400)
sub-1_ses-20230226_ecephys	1	F	P8W/P12 W	0.36	1020	1528	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(976)
sub-2_ses-20230902_ecephys	2	F	P8W/P12 W	0.13	708	1492	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(952)
sub-3_ses-20230218_ecephys	3	F	P8W/P12 W	0.08	431	1531	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(400)
sub-3_ses-20230225_ecephys	3	F	P8W/P12 W	0.28	917	1532	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(992)
sub-4_ses-20230907_ecephys	4	F	P8W/P12 W	0.24	885	1483	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(960)
sub-5_ses-20230706_ecephys	5	F	P8W/P12 W	0.02	154	764	ElectrodeGroup0, ElectrodeGroup3	Stimulus presentations(960)
sub-5_ses-20230822_ecephys	5	F	P8W/P12 W	0.13	664	1354	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(952)
sub-6_ses-20230823_ecephys	6	F	P8W/P12 W	0.14	591	1413	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(952)
sub-7_ses-20220831_ecephys	7	F	P8W/P12 W	0.30	928	1530	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(416)
sub-8_ses-20220228_ecephys	8	F	P8W/P12 W	0.34	851	1529	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(320)
sub-9_ses-20230908_ecephys	9	F	P8W/P12 W	0.19	880	1479	ElectrodeGroup0, ElectrodeGroup1, ElectrodeGroup2, ElectrodeGroup3	Stimulus presentations(960)

Other data in each file (not exported)

File (session_id)	Acquisition	Processing modules: interfaces
sub-0_ses-20230901_ecephys	Ketamine infusion start	
sub-1_ses-20230218_ecephys	Ketamine infusion start	
sub-1_ses-20230226_ecephys	Ketamine infusion start	
sub-2_ses-20230902_ecephys	Ketamine infusion start	
sub-3_ses-20230218_ecephys	Ketamine infusion start	
sub-3_ses-20230225_ecephys	Ketamine infusion start	
sub-4_ses-20230907_ecephys	Ketamine infusion start	
sub-5_ses-20230706_ecephys	Ketamine infusion start	
sub-5_ses-20230822_ecephys	Ketamine infusion start	
sub-6_ses-20230823_ecephys	Ketamine infusion start	
sub-7_ses-20220831_ecephys	Ketamine infusion start	
sub-8_ses-20220228_ecephys	Ketamine infusion start	
sub-9_ses-20230908_ecephys	Ketamine infusion start	

4. Units per region (Computed by this export)



5. All region labels (Computed by this export)

One row per region label exactly as stored in the NWB files, all units (no quality filter), sorted by unit count. Firing-rate columns summarise the NWB units.firing_rate values where the dataset provides them.

Region label	Units	Files	Subjects
NA	867	13	10
LSr	521	7	7
CP	516	7	5
ORBI5	360	5	5
PO	345	5	5
ACB	317	8	7
MB	310	8	7
OLF	277	9	8
RSPv5	243	6	5
MD	240	4	4
PIR	212	3	3
ZI	201	7	7
Alv5	171	2	2
PAG	159	5	4
ACAd5	153	6	6
Ald5	153	2	2
CA1	146	7	6
EPd	135	1	1
PL5	128	3	3
ORBI2/3	118	4	4
MOp6a	113	4	4
ORBvl2/3	111	2	2
VPM	107	1	1
NPC	106	6	6
LP	102	5	5
MRN	100	2	2
ORBI6a	100	3	3
PoT	95	2	2
MOS6a	91	4	4
OT	91	2	2
AON	89	2	2
RSPv6a	89	3	2
DG-mo	87	6	6
PL2/3	85	1	1
SI	79	4	3
MOp5	79	3	3
ACAv2/3	77	4	4
ACAv5	77	3	3
Ald6a	70	2	2
SUB	67	3	3

Region label	Units	Files	Subjects
LSv	64	5	5
TH	62	6	6
FF	59	4	4
STR	59	9	7
HY	59	7	7
ORBvl5	57	2	2
CA3	57	6	6
VPLpc	52	2	2
RSPv2/3	51	5	4
ORBm5	50	1	1
SCig	50	2	2
PL1	49	1	1
VPMpc	49	2	2
RT	46	1	1
ILA1	46	1	1
ACAv1	43	4	4
VAL	40	2	2
SF	40	1	1
SSp-m6a	40	1	1
MOs5	39	5	5
ACA2/3	39	4	4
DP	37	3	3
Alv2/3	36	2	2
BST	35	3	3
SSp-II6a	35	1	1
RSPv1	35	5	4
NLOT2	32	1	1
DG-sg	29	7	6
LGv	28	3	3
COApm	26	2	2
ILA5	26	1	1
SCiw	25	2	1
RSPd6a	25	2	2
PSTN	25	2	2
LHA	25	2	2
PCN	23	1	1
AAA	23	1	1
FRP6a	22	1	1
ORBI1	22	1	1
MOp2/3	21	3	3
RSPagI5	20	1	1
CM	20	2	2
OP	20	2	2

Region label	Units	Files	Subjects
STN	19	1	1
ORBm6a	18	1	1
LH	18	3	3
SSp-m5	17	1	1
Eth	16	1	1
SSp-m4	16	1	1
PA	15	1	1
IGL	15	2	2
SCsg	15	2	1
TTd	14	3	3
LGd-sh	13	1	1
CEAm	13	1	1
IG	13	1	1
PRC	13	2	2
LGd-ip	12	1	1
RSPv6b	12	3	2
LGd-co	12	3	3
NOT	11	2	2
ILA2/3	11	1	1
RPF	11	1	1
RSPd5	11	3	3
PL6a	11	1	1
SCdg	10	1	1
DG-po	10	4	4
FS	10	1	1
MOp6b	10	3	3
ACAAd1	9	1	1
SCzo	9	3	2
HPF	9	3	3
CLI	9	1	1
SSp-ul6a	8	1	1
SCop	8	2	1
SPA	8	2	2
LSc	8	1	1
SCdw	8	1	1
SPFp	8	1	1
RSPagl2/3	8	1	1
PPT	7	1	1
ND	7	1	1
RSPagl6a	7	1	1
SSp-m2/3	7	1	1
SNC	6	1	1
CA2	5	2	2

Region label	Units	Files	Subjects
ACAv6a	4	1	1
IF	4	1	1
SSp-m1	4	1	1
ACAd6b	3	1	1
SSp-II6b	3	1	1
SNr	3	1	1
RSPd2/3	3	1	1
MGm	3	1	1
PS	3	1	1
RSPagl6b	2	1	1
NLOT1	2	1	1
NLOT3	2	1	1
PF	2	1	1
BLAv	1	1	1
BMAp	1	1	1
BLAa	1	1	1
MEPO	1	1	1
MGv	1	1	1
MGd	1	1	1
IMD	1	1	1
COApI	1	1	1
PVT	1	1	1
SSp-ul6b	1	1	1
SSs6b	1	1	1

6. Intervals tables (stimuli, trials, epochs)

Each NWB intervals table, with the number of files containing it, total rows, and its columns as stored.

Table	Files	Rows	Columns
Stimulus_presentations	13	10,192	start_time, stop_time, stimulus

7. Column and signal descriptions from the NWB files

Quoted verbatim from sub-0_ses-20230901_ecephys.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	x	the x coordinate of the position (+x is posterior)
electrodes	y	the y coordinate of the position (+y is inferior)
electrodes	z	the z coordinate of the position (+z is right)
electrodes	imp	the impedance of the electrode, in ohms
electrodes	filtering	description of hardware filtering, including the filter name and frequency cutoffs
units	(table)	Autogenerated by NWBFile
units	snr	Signal-to-noise ratio
units	n_spikes	Number of spikes in the entire recording
units	quality	Quality label from Bombcell pipeline
units	channel	Closest electrode channel id
units	brain_region	Assigned AIBS CCFv3 brain region acronym
units	x	x location in AIBS CCFv3 brain atlas
units	y	y location in AIBS CCFv3 brain atlas
units	z	z location in AIBS CCFv3 brain atlas
units	spike_times	the spike times for each unit
intervals/Stimulus presentations	(table)	experimental intervals
intervals/Stimulus presentations	start_time	Start time of epoch, in seconds
intervals/Stimulus presentations	stop_time	Stop time of epoch, in seconds
intervals/Stimulus presentations	stimulus	Type of stimulus delivered
acquisition	Ketamine infusion start	Approximate timepoint of manual infusion of ketamine bolus

8. Exported files

File	Rows	Columns
tables/sessions.csv	13	session_id, asset_path, size_gb, subject_id, species, sex, age, genotype, strain, session_start_time, session_description, institution, lab, n_units, n_electrodes, electrode_groups, region_source, regions, intervals, acquisition, processing
tables/units.parquet	9,554	session_id, unit_id, unit_row, region, snr, n_spikes, quality, channel, brain_region, x, y, z, subject_id
tables/region_summary.csv	150	region, n_units, n_sessions, n_subjects
tables/intervals_summary.csv	1	table, n_sessions, n_rows, columns
intervals/<table>.parquet	one per intervals table	session_id + the table's columns
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		published metadata quoted in sections 1 and 7